Research questions from the current analysis

Multimodal drivers + TerraMind embeddings for daily soil moisture

Prajwal Khanal

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0. Overview

Eight research questions follow from the results we already have. For each one: the question, the evidence in hand, and what is still missing. Section 2 then works through RQ1 in depth, using only the results that exist today.

Block	RQs	Role in the thesis / paper
Core	RQ1–RQ3	Does it work, what does it generalise to, where does it fail
Mechanism	RQ4–RQ5	Which inputs matter at which depth; is the fine path real
Negative	RQ6	LST as an auxiliary signal — a clean negative result
Outlook	RQ7–RQ8	Management signals (irrigation); transfer to an unseen region

1. The eight research questions

RQ1 Do frozen geospatial-foundation-model embeddings add skill beyond a multimodal driver baseline?

Does fusing TerraMind tokens (S2, S1, DEM, LULC) with ERA5-Land, SIF, TWSA and soil data improve daily soil moisture at 0–10, 10–30 and 30–100 cm over a drivers-only model?

- **Have:** modality-shuffle ablation of the §59 model on 221 OOS stations (Section 2); TerraMind-value figure.
- **Missing:** a drivers-only model trained to 10 epochs; a raw-imagery control; the within-station S2 shuffle; the UMAP of the embeddings (deferred).

RQ2 What does the model generalise to — places or processes?

How does skill degrade from random validation to out-of-time (2023–2025) to out-of-sample stations, and does the model memorise site fingerprints instead of learning soil-moisture dynamics?

- **Have:** splits §47 (train 573 / val 89 / OOS 224, OOT 2023–2025). Patchwise `pw_stage2a_L3`: OOT 0.0434 beat val 0.0507 at 0–10 cm (verdict: memorisation). Step-0 offsets probe: pair sign 0.50 (no-go); site-fingerprint verdict confirmed by rerun 27572594.
- **Missing:** one table with val / OOT / OOS for the selected model, per depth.
- Probably the strongest and most honest finding of the thesis.

RQ3 How does skill vary across the landscape?

How does prediction error vary with land cover, climate/aridity, elevation, terrain and soil texture, and with depth?

- **Have:** per-station metrics; LULC, elevation band, soil, HAND for every station. HAND is stable under DEM noise, TWI is not (dTWI goes negative under 0.2 m noise) — use HAND.
- **Missing:** the join of per-station metrics to the landscape descriptors and the stratified plots. Note the elevation sentinels (18 REMEDHUS = -99.9, Baytik = 0.0) must be flagged first.

RQ4 Which modalities matter, at which depth?

What is the relative contribution of S2, S1, DEM/LULC, ERA5, SIF, TWSA and soil to surface versus root-zone soil moisture?

- **Have:** the ablation table (Section 2): ERA5 dynamics dominate the surface and fade with depth; soil properties grow with depth (+7.6 % $\rightarrow$ +25.7 %); TWSA ≈ 0; SIF ≈ 1.5 %. §58 180-day ERA5 window run.
- **Hypothesis:** optical/SAR drive the surface, TWSA and ERA5 memory drive the deep layer — the TWSA part is *not* supported so far.

RQ5 Does the fine-resolution path carry real sub-kilometre information?

Does the predicted spatial pattern reflect within-tile soil-moisture heterogeneity, or does it flatten?

- **Have:** §53 fine ablation — soil moisture ignores the 20 m path (LST reads it); fine-path shuffle costs only +0.1 to +1.3 % ubRMSE; the §48 map flattens after epoch 8; TxSON effective-resolution test (§70); Overijssel grid run (§68, paused).
- A publishable negative result if framed well.

RQ6 Is land-surface temperature a useful auxiliary signal for soil moisture?

Does adding LST as input or as auxiliary supervision help soil-moisture prediction?

- **Have (all negative):** the within-tile Landsat LST pattern is static (+0.967 across months); the pooled correlation is Simpson's paradox (+0.167 pooled, -0.077 within station); ECOSTRESS DTR is the daytime pattern ($r = +0.816$), so G0 fails; in training LST hurts SM (best val 0.0537 with LST vs 0.0487 without).
- Ready to write as a short paper or section.

RQ7 Can the model detect management signals?

Can a GFM-based model detect anthropogenic soil-moisture signals such as irrigation that the weather drivers do not explain?

- **Have:** Mead Ne1/Ne2/Ne3 daily SM (§69); irrigation shows only in the dry year (2022).
- **Missing:** the irrigation log covers 2014–2021 only; prediction-minus-observation analysis.

RQ8 Does the representation transfer to an unseen region?

Do predictions hold at full grid scale in a region never seen in training?

- **Have:** the Overijssel run (§68: cloud mask done 688/688; prep, tokenise, predict pending); the 8 TWENTE stations, all OOS (end 2020, so no OOT).

2. RQ1 in depth: what can we answer *today*?

In one sentence: With current results we cannot claim that the GFM adds skill, but we can say what the model uses it for: a static landscape descriptor that conditions the weather response, not a dynamic observation of soil moisture.

2.1 The evidence: the modality-shuffle ablation

Model: §59 (3 layers, no LST, warmup 200). Data: OOS split, paired per-station comparison (`eval_output/baseline_se`
Baseline ubRMSE is 0.0500 / 0.0479 / 0.0476 m^3m^{-3} and r is 0.766 / 0.761 / 0.702 at 0–10 / 10–30 / 30–100 cm (n = 221 / 144 / 119 stations).

Cross-station swaps another station's input in (destroys site identity); *within-station* shuffles the dates of the same station (keeps identity, destroys timing).

Table 1: Change in ubRMSE (%) when one input is shuffled, with the ablated r in brackets and the share of stations made worse. Baseline r: 0.766 / 0.761 / 0.702.

	0–10 cm		10–30 cm		30–100 cm	
Shuffled input	ΔubRMSE (r)	worse	ΔubRMSE (r)	worse	ΔubRMSE (r)	worse
Satellite / GFM tokens						
All satellite, cross	+36.8 (0.52)	97%	+37.1 (0.52)	96%	+38.4 (0.44)	92%
S2, cross	+4.9 (0.74)	84%	+6.8 (0.72)	87%	+8.7 (0.63)	85%
S1, cross	+4.8 (0.73)	82%	+7.7 (0.70)	84%	+14.4 (0.58)	86%
S1, within	0.0 (0.75)	49%	+1.0 (0.73)	59%	0.0 (0.64)	49%
DEM, cross	+12.6 (0.71)	90%	+9.7 (0.68)	86%	+18.4 (0.59)	96%
LULC, cross	+6.1 (0.73)	82%	+6.6 (0.70)	82%	+12.7 (0.62)	87%
Fine 20 m path, cross	+0.5 (0.75)	56%	+1.3 (0.74)	65%	+0.1 (0.68)	52%
Drivers and tabular						
ERA5, cross	+30.3 (0.50)	95%	+24.6 (0.54)	91%	+15.1 (0.50)	86%
ERA5, within	+39.0 (0.35)	88%	+30.4 (0.37)	81%	+13.5 (0.34)	73%
Soil, cross	+7.6 (0.72)	87%	+17.6 (0.63)	92%	+25.7 (0.55)	94%
SIF, cross	+1.3 (0.75)	63%	+1.6 (0.73)	62%	+1.8 (0.67)	71%
TWSA, cross	0.0 (0.77)	50%	0.0 (0.76)	53%	+0.1 (0.71)	58%

2.2 What we can claim now

1. **The model relies on the GFM tokens as much as on the weather.** Swapping all satellite tokens between stations raises ubRMSE by about 37 % at every depth, makes 92–97 % of stations worse, and drops r from about 0.77 to about 0.52 — the same order as shuffling ERA5.
2. **The tokens act as a static landscape descriptor, not a dynamic observation.** Shuffling S1 dates *within* a station costs nothing (0 to +1 %, about half the stations worse — i.e. noise). The day-to-day signal comes from ERA5: its within-station shuffle is the single largest loss at the surface (+39 %) and fades with depth, as physics expects.

3. **The tokens shape the response, not just the mean.** ubRMSE removes the bias, so a 37 % loss means the tokens set the amplitude and shape of each site's response to the weather. Framing: *GFM embeddings condition the weather→soil-moisture transfer on the landscape.*
4. **The depth pattern is inverted relative to a moisture signal.** If the tokens carried SAR/optical moisture information their effect would peak at the surface. Instead S1 reliance grows from +4.8 % (0–10 cm) to +14.4 % (30–100 cm), the same direction as the static soil properties — again consistent with site description.
5. **The satellite modalities are partly redundant.** The individual S2 + S1 + DEM + LULC effects sum to about 28 % at 0–10 cm, less than the 37 % for shuffling them together: they encode overlapping site information.
6. **The fine 20 m path and TWSA are unused** ($\leq$1.3 % and $\approx$0 %).

2.3 What we cannot claim yet

- **Added skill.** A shuffle measures *reliance* of a trained model, not *added value*: a drivers-only model might learn the same site information from soil and coordinates. In the text, write "reliance" or "use", never "added value".
- **GFM versus raw imagery.** Nothing yet separates TerraMind from the pixels it was given.
- **S2 dynamics.** Only the cross-station S2 shuffle exists; the within-station S2 shuffle has not been run.

2.4 How to answer RQ1 fully (later)

All runs at 10 epochs, compared at epoch 10, OOS and OOT splits. Every job needs an explicit OK.

1. **Within-station S2 and all-satellite shuffle** on the §59 checkpoint — eval only, cheap.
2. **Run A — drivers + soil only, no satellite tokens.** The core counterfactual. Needs a new `--drop-modalities` flag in `train.py`.
3. **Run B — Run A + static site descriptors** (LULC class, elevation, HAND, soil). Tests whether TerraMind is an expensive station identifier.
4. **Optional:** raw S2 bands/indices + S1 VV/VH at the station pixel in place of the tokens (foundation model vs imagery).

Metrics per depth: ubRMSE, r, and anomaly-r (seasonal climatology removed — the metric that shows dynamics); paired per-station deltas with bootstrap CIs (reuse the ablation-summary machinery); stratified by land cover and aridity (feeds RQ3).

Table 2: How to read the retrained controls.

Outcome	Interpretation
Full $>$ B $>$ A, gain in anomaly-r	GFM adds *dynamic* skill
Full $\approx$ B $>$ A	GFM adds only static site information (what Section 2.2 predicts)
Full $\approx$ A	No added skill — still a publishable answer